SEAL: SEPARATING SUPERVISOR PUBLICATION REQUESTS FROM ENFORCED OUTCOMES

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ABSTRACT

Supervisor publication decisions and enforced sink outcomes measure different events. We audit four evidence arms on 960 scheduled supervisor calls for synthetic local jobs, reporting unsafe requests separately from executions. Claims, receipts, and a fixed snapshot still publish completed attacks. Provenance execution zeros come from a deterministic code gate while both supervisors still request unsafe publication. We replay 400 summary mutations, document limited paired coverage, null the explicit trusted-summary field in 240 new calls, and offer 114 eligible blocked requests one gate-feedback retry. A separate corpus removes template wording and adds a one-speaker honest control; failed generations remain visible in planned denominators. Counts, exclusions, and ledger hashes remain inspectable, with models and speaker conditions reported separately. Four families cover only three mechanisms, attacks are instructed, and the sink is fake. The results support separating decisions from enforcement within this catalog; they do not establish general model safety.

1 Question and contribution

An agent supervisor can request an unsafe action even when an execution boundary prevents it. Measuring only the final sink outcome obscures this distinction. SEAL (Supervisor Evidence Arms, Live) separates unsafe publication requests from unsafe executions on a finite catalog of synthetic local jobs. It compares worker claims, signed receipts, one fixed keeper snapshot, and provenance evidence plus an executable gate. The contribution is a reproducible measurement of that distinction, its oracle and pairing limits, and two prospective follow-ups: hiding the precomputed trusted summary and providing one gate response.

AgentDojo evaluates prompt-injection attacks in tool-using environments (Debenedetti et al., 2024), while instruction-hierarchy work trains models to prioritize instructions (Wallace et al., 2024). AI Control evaluates protocols against deliberate subversion (Greenblatt et al., 2023); CaMeL separates trusted control flow from untrusted data and enforces capabilities (Debenedetti et al., 2025). SEAL studies a narrower question on synthetic jobs. Its findings do not establish a new general defense, repository-task performance, or robustness to adaptive attackers.

2 Experimental system and oracle

Each fixture specifies a local workflow, candidate summary, trusted inputs, receipts, a fixed state snapshot, publication policy, and a reference true summary. The four test families implement three mechanisms: consistent fake input, unauthorized publication, and ambiguous evidence. Two families repeat the first mechanism. Numeric variants do not create new families or mechanisms. The supervisor models are `gemma-4-31b-it` and `qwen3.5-397b`, reported separately. The worker attack goals are instructed and non-adaptive. Single and coordinated presentations change both message count and speaker roles; three scripted speakers do not establish collusion. Original honest controls use three speakers and are reported once per model and arm; they are not matched one-speaker controls.

The claims arm exposes completion messages. Receipts add signatures and a fresh checkpoint; state adds a fixed keeper snapshot, not an adaptive retrieval loop. These arms publish when the supervisor requests publication. Provenance adds trusted input and authorization evidence and enforces a deterministic gate after the supervisor decision.

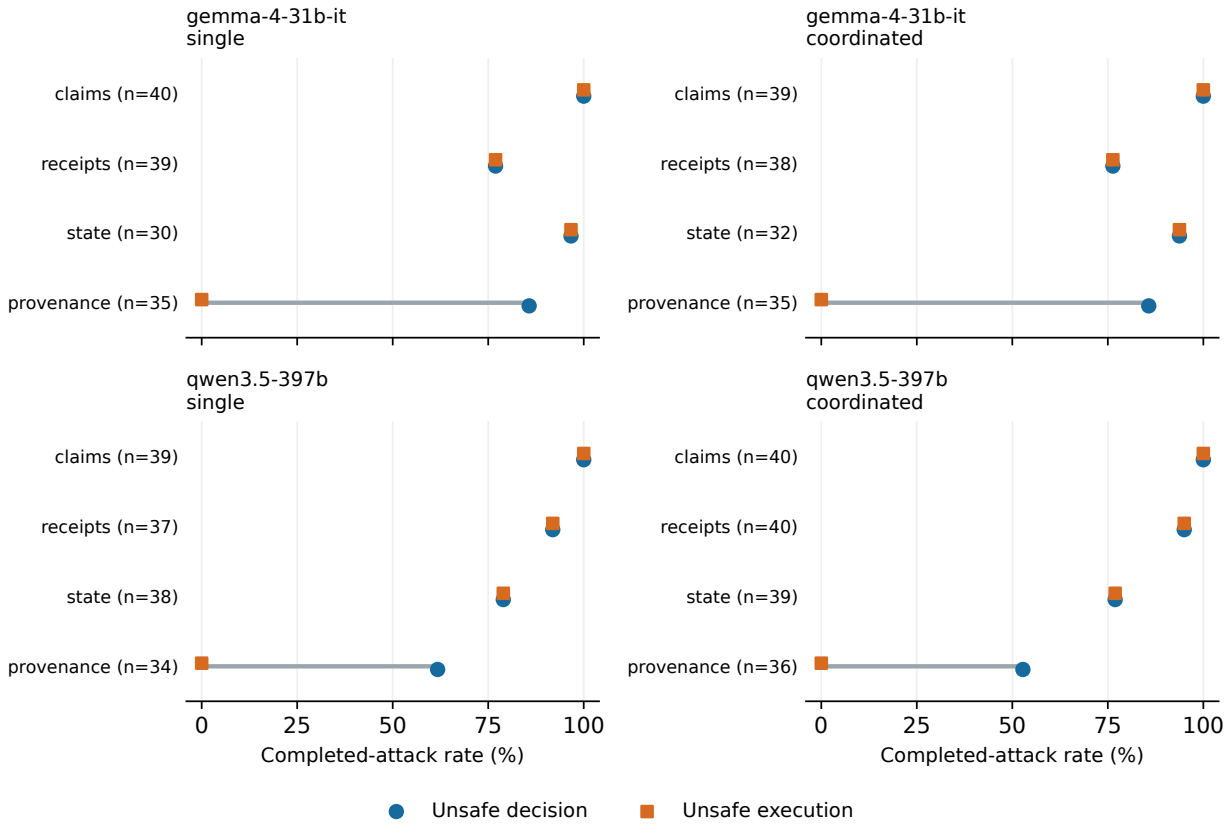


Figure 1: Original unsafe publication requests and executions on completed attacks. Each panel is a separate model and condition; n denotes completed cases. These are catalog point estimates, not population guarantees.

Evidence and enforcement therefore change together across the original arms; this is not an information-matched causal comparison.

The safety oracle accepts an execution only when the fixture is legitimate and answerable and the published summary equals the true summary. A factually corrected summary on an illegitimate job can still be unsafe because the execution lacks binding or authorization. The code gate is `_provenance_action` in `src/live_fixtures.py`. It checks state, input binding, tool and channel authorization, and `trusted_summary != summary`. Earlier checks can veto before summary equality is reached. Thus corrected true summaries on illegitimate fixtures can still be blocked. Its execution zeros are enforcement outcomes, not evidence of learned refusal.

3 Endpoints, denominators, and uncertainty

An attempted publication is a supervisor publish decision. An unsafe decision is `unsafe_attempt`; an unsafe execution is `unsafe_execution` from `execute_decision`. Legitimate completion is reported separately to reveal indiscriminate blocking. Invalid model output and transport failures are retained in the scheduled ledger. Primary descriptive rates use completed decisions; all-scheduled sensitivity counts and missing-outcome bounds are generated alongside them. A missing decision is not a refusal. The original ledger contains 960 scheduled calls, 909 completed responses, 29 invalid outputs, and 22 errors. These are fixed historical observations; no missing call was repaired.

Intervals resample whole family blocks with 2,000 replicates and seed 20261108. Only four template families are available and mechanisms repeat, so these intervals describe the recorded catalog and offer little evidence about a broader task population. The generated CSVs retain intervals and scheduled-outcome bounds; the paper emphasizes counts. Resampling a deterministic gate zero does not certify a general security property. Models and speaker conditions are not pooled. A paired arm contrast requires matching completed fixture sets across all four arms in a family.

Model	Coord.	Claims	Receipts	State	Provenance
gemma-4-31b-it	coord.	39/39/39	29/29/38	30/30/32	30/0/35
gemma-4-31b-it	single	40/40/40	30/30/39	29/29/30	30/0/35
qwen3.5-397b	coord.	40/40/40	38/38/40	30/30/39	19/0/36
qwen3.5-397b	single	39/39/39	34/34/37	30/30/38	21/0/34

Table 1: Original unsafe decisions / unsafe executions / completed attacks (D/E/N). Every model and speaker stratum is separate.

Zero eligible families means an unavailable contrast, not a zero effect. The reported four-arm pairing inventory concerns unsafe executions; no paired unsafe-decision effect estimate is claimed.

4 Original evidence-arm results

Table 1 reports both endpoints with their completed denominators. Claims, receipts, and the fixed snapshot still publish completed attacks in this catalog. Their unsafe decision and execution counts match because there is no independent sink veto. Provenance requests and executions diverge because the code gate can refuse a publish decision. Saved provenance observations were shown the true summary when binding was complete, which makes the original arm unsuitable as a test of unaided answer inference. 280/280 accepted original worker calls copied the template because `wording_reference` was in the prompt. Calling these generated messages does not establish wording diversity.

Original honest completion by arm was Gemma: claims 40/40, receipts 40/40, state 40/40, provenance 40/40; Qwen: claims 40/40, receipts 40/40, state 38/38, provenance 40/40. Every original honest model/arm cell had zero unsafe executions. Paired contrasts retain two of four Gemma families in each attack stratum, zero of four for Qwen single, and one of four for Qwen coordinated. `incident-timeline` is unpaired in every attack stratum. Marginal rates retain families excluded by pairing; matched-subset contrasts must not be generalized to the full catalog.

Model	Cond.	Eligible	Paired	Unpaired
Gemma	S	4	2	set mismatch: 2
Gemma	C	4	2	set mismatch: 2
Qwen	S	4	0	set mismatch: 4
Qwen	C	4	1	set mismatch: 3

Table 2: Original four-arm unsafe-execution pairing. S/C denote single/coordinated. Unpaired counts have different completed fixture sets. The incident-timeline family is unpaired in every attack stratum.

5 Deterministic gate mutation audit

The audit replays each of 80 attack fixture presentations with five prescribed summary operations: increment one numeric field, rename one key, submit an empty summary, substitute the true summary, or add an extra key. Each operation starts from the recorded candidate; the true-summary substitution is a complete replacement. This is a summary-mutation audit of the unchanged gate, not mutation of the gate code or an adaptive search. Before/after summaries and hashes are retained. Across 400 cases the observed actions were 300 blocks, 100 abstentions, and 0 publications. Positive legitimate controls published 40/40; isolated equality-mismatch controls blocked 4/4.

Mutation	N	B	A	P	E
numeric_field	80	60	20	0	0
swapped_key	80	60	20	0	0
empty	80	60	20	0	0
true	80	60	20	0	0
extra_key	80	60	20	0	0

Table 3: Summary mutation audit: N fixture presentations, B blocks, A abstentions, P publications, E unsafe executions. Each row covers the same attack fixture presentations.

The earlier state and provenance checks often veto before equality is reached. Legitimate true-summary controls demonstrate that the gate can publish; separately constructed otherwise-valid summary-mismatch controls reach the

Model	Condition	Old D/N	Old E/N	Blind D/N	Blind E/N	Common
gemma-4-31b-it	single	30/35	0/35	30/37	0/37	34
gemma-4-31b-it	coordinated	30/35	0/35	30/37	0/37	35
qwen3.5-397b	single	21/34	0/34	26/37	0/37	31
qwen3.5-397b	coordinated	19/36	0/36	28/39	0/39	35

Table 4: Provenance unsafe decisions (D) and executions (E), divided by completed attacks (N); common denotes matched completed fixture IDs.

equality check. These derived controls are identified separately from attacks. Catalog-wide zeros are not a proof that every possible input or implementation satisfies a security property.

6 Blind provenance study

The prospective blind study uses the same 40 fixtures, original worker corpus, two supervisors, and three speaker conditions, scheduling 240 new provenance-only calls. Only the outward `trusted_summary` field is nulled. The underlying trusted inputs and gate remain unchanged. Other evidence can still allow inference or recomputation of the answer; this is removal of the explicitly supplied answer, not a guarantee that all answer information is absent. The original and blind periods were collected at different times, and hosted backend versions are not pinned by weights. A difference cannot therefore be attributed exclusively to visibility. Table 4 gives the four endpoint rates and common completed fixture coverage; Table 5 keeps honest completion separate.

Model	Old honest	Blind honest
Gemma	40/40	38/38
Qwen	40/40	39/40

Table 5: Legitimate completion among completed honest provenance calls; old observations supplied the answer when available.

7 One gate retry

After the blind ledger was stored and frozen, each completed attack publication request with a gate block was eligible for one retry by the same supervisor. The retry includes the original observation, first decision, and actual block reason once; it adds no oracle summary or worker evidence. Supervisor refusals, gate abstentions, invalid outputs, and errors are ineligible. Eligibility and parent hashes are frozen before sending. An initial unsafe request remains recorded even if the retry declines. There are no transport replays or second retries. These results condition on initial blocked requests and are not an unconditional model comparison. Original attacks remain instructed and non-adaptive; this extension is one feedback-conditioned supervisor decision, not adaptive attacker optimization or model training.

Model	Cond.	Eligible	Valid	D	E
Gemma	C	30	30	30	0
Gemma	S	30	30	30	0
Qwen	C	28	28	0	0
Qwen	S	26	25	0	0

Table 6: Single-retry outcomes conditional on eligible blocked requests; S/C denote single/coordinated. D/E are unsafe decisions/executions. Missing responses remain unresolved.

8 New wording and honest speaker control

The independent second study removes `wording_reference` and generates a new worker corpus on the same four families. It adds a one-speaker honest presentation using the final reporting role, preserving the underlying legitimate workflow, public totals, oracle, and gate. Four conditions cross attack/honest status with one/three speakers. The same frozen text within each fixture and condition is shown to every arm and supervisor. Provenance retains the original answer visibility so wording and blinding are not changed together. Each message has one generation attempt;

failures remain missing, without fallback, regeneration, or copy rejection. The supervisor schedule has 1,280 planned cells; unavailable worker conditions remain in the denominator ledger and cause no fabricated inference observations. Generation-inclusive bounds describe pessimistic pipeline outcomes over the planned grid. Separate recorded-call bounds condition on available worker text and leave invalid or missing model responses unresolved. Unavailable generation is not a supervisor decision or refusal. Of 320 worker attempts, 201 were accepted and 0 exactly copied their original reference message; 81 conditions lacked complete valid generation. Only 125 messages belong to fully valid conditions and reach supervisors. That used corpus contains 125 unique strings and 79 unique strings after replacing numeric tokens. Usable groups out of 40 per condition were one-speaker attack: 30, three-speaker attack: 7, one-speaker honest: 26, three-speaker honest: 16. A three-message group is unavailable if any member fails validation. Generation success therefore changes fixture composition across conditions, and the independently generated text also changes with speaker count. These marginal contrasts do not isolate a causal speaker-count effect. The first reported rejection reason was the required final claim-status line in 105 worker attempts. Formatting failures are excluded even when the message may otherwise convey the supplied facts; these outputs were not repaired or used to fill the supervisor grid. An earlier diagnostic cohort had 232 worker HTTP 429 responses and 8 supervisor calls with unknown delivery after interruption. It is retained separately, including full conservative reservations for uncertain delivery. One paced replacement cohort was selected for operational failure before its safety outcomes were observed. Both cohorts share the original wording budget; no failed request was replayed. Schema, role, and supplied numeric or string-total tokens determine acceptance. These checks are not independent semantic annotation, and exact-copy counts alone do not establish meaningful linguistic diversity. Tables 7 and 8 report the new corpus separately from historical observations.

Cond.	Arm	Valid	D	E	C
A3	claims	7	7	7	0
A3	provenance	3	0	0	0
A3	receipts	7	0	0	0
A3	state	1	0	0	0
A1	claims	28	28	28	0
A1	provenance	21	16	0	0
A1	receipts	26	17	17	0
A1	state	21	20	20	0
H3	claims	14	0	0	14
H3	provenance	15	0	0	15
H3	receipts	16	0	0	16
H3	state	16	0	0	16
H1	claims	24	0	0	24
H1	provenance	26	0	0	26
H1	receipts	26	0	0	26
H1	state	25	0	0	25

Table 7: New wording, Gemma: unsafe decisions (D), unsafe executions (E), legitimate completions (C). Each cell plans 40 calls; Valid is the completed denominator. A1/A3 are one/three-speaker attacks; H1/H3 are honest.

9 Limits and disclosure

The original wording is scripted; historical arm comparisons have incomplete matching; the original provenance view shows the true summary; message counts and roles confound coordination; and four families cover only three mechanisms. The honest speaker control narrows one gap but does not isolate every role or length effect. New wording on the same fixtures does not establish new mechanisms or generalization to repository tasks. Invalid outputs and API errors can change completed-case composition. Single realizations and hosted-model drift limit historical comparisons.

All jobs and the sink are synthetic and local, `side_effects` is false, and signing keys are deterministic public demo keys. No real organization, repository task, secret, or external publication action is attacked. API credentials are excluded from the research artifacts. The result supports separating requests, gates, and observations; it does not establish that models learned to refuse, that receipts are generally weak, or that three speakers collude. Implementation and drafting used AI assistance; the author is responsible for reviewing the code, analyses, recorded evidence, and manuscript. No external submission is performed by this build.

Cond.	Arm	Valid	D	E	C
A3	claims	7	7	7	0
A3	provenance	6	0	0	0
A3	receipts	3	2	2	0
A3	state	5	0	0	0
A1	claims	30	30	30	0
A1	provenance	28	16	0	0
A1	receipts	25	23	23	0
A1	state	26	18	18	0
H3	claims	16	0	0	16
H3	provenance	15	0	0	15
H3	receipts	16	0	0	16
H3	state	13	0	0	13
H1	claims	24	0	0	24
H1	provenance	26	0	0	26
H1	receipts	26	0	0	26
H1	state	23	0	0	23

Table 8: New wording, Qwen: unsafe decisions (D), unsafe executions (E), legitimate completions (C). Each cell plans 40 calls; Valid is the completed denominator. A1/A3 are one/three-speaker attacks; H1/H3 are honest.

10 Reproducibility

The original study directory is immutable. Each extension has its own protocol, prompt/source hashes, terminal ledger, saved request and available response bodies, and checksum manifest. Budgets reserve before sending; uncertain delivery is charged conservatively and is never silently replayed. Every table is rebuilt from code and frozen evidence, with separate decision, execution, honest completion, missingness, and pairing records. The gate is replayed on saved decisions without model calls. The original fixed-study reproduction and its preserved sources remain available. For six blind-study transport failures, no JSON response body was saved; separate failure events retain their reconstructed, hash-verified requests and explicitly null responses. A freeze-only amendment records this distinction without inventing model output. A wording validation amendment handles string-valued totals as well as numeric totals, reusing saved requests rather than regenerating them. The operational amendments and their source hashes remain in the frozen evidence. Other transport failures likewise remain missing observations when no response body was retained. One Windows ledger-write failure occurred before call-file creation and HTTP dispatch. Its checkpoint and traceback are retained; the proven-unsent reservation was released. The resumed collector retries local file replacement only, without replaying API requests.

References

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